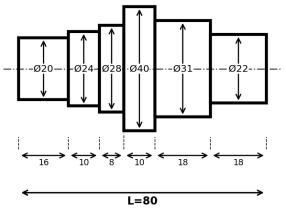
HOJA DE PROCESO
1ºME - TRINIDAD ARROYO

CROQUIS DE LA PIEZA:



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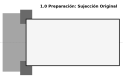
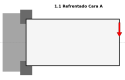
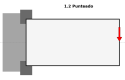
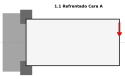
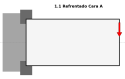
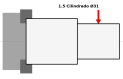
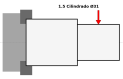
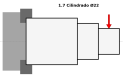
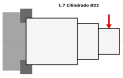
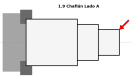
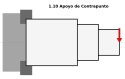
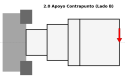
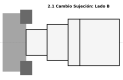
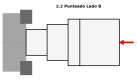
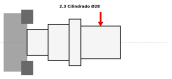
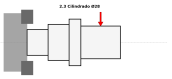
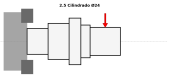
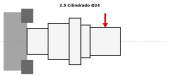
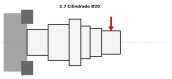
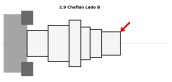
Alumno: Sergio Abad | **Material:** Acero AISI 1045 | **Bruto:** Ø50 x 140 mm | (Rev. A)

Nº Op.	Operación	Herramienta	Vc (m/min)	N (rpm)	a(mm/rev)	P (mm)	Notas / Medic clave	Croquis
1.0	Colocar tocho y alinear	—	—	—	—	—	Asegu conce inicial	
1.1	Refrentar cara frontal	Plaquette torneado derecha	120–160	800–1000	0.15–0.25	0.5–1	Dejar longitu total ≈ +2 mm sobre	
1.2	Punteado	Broca centrar Ø4-5	15–20	400–600	Man.	3–5	Avanc lento; cono 60º; refrige	
1.3	Cilindrar Ø40 (desbaste)	Plaquette desbaste	100–140	600–800	0.3–0.5	2–3 por pasada	Hasta Ø41–42 mm	
1.4	Cilindrar Ø40 (acabado)	Plaquette acabado	140–180	1000–1200	0.15–0.25	0.3–0.5	Ø40 mm final	
1.5	Cilindrar Ø31 (desbaste)	Plaquette desbaste	100–130	700–900	0.3–0.5	2–3	Longit aprox. 36 mm + marge	
1.6	Cilindrar Ø31 (acabado)	Plaquette acabado	130–170	1000–1300	0.15–0.20	0.3	Ø31 mm	
1.7	Cilindrar Ø22 (desbaste)	Plaquette desbaste	90–120	800–1000	0.25–0.4	2	Longit aprox. 18 mm + marge	
1.8	Cilindrar Ø22 (acabado)	Plaquette acabado	120–160	1100–1400	0.12–0.20	0.3	Ø22 mm	
1.9	Chaflanes / redondeos	Plaquette o herramienta chaflán 45°	100–140	800–1000	manual	—	Chafla 1×45° en todos los escalo	
1.10	Colocar contrapunto (si no estaba)	Contrapunto fijo o giratorio	—	—	—	—	Apoya en el punto centra ajusta presió moder	

Nº Op.	Operación	Herramienta	Vc (m/min)	N (rpm)	a(mm/rev)	P (mm)	Notas / Medic clave	Croquis
2.0	Colocar contrapunto en el punto existente	Contrapunto	—	—	—	—	Verific alinea y presió	
2.1	Refrentar cara final	Plaquette torneado	100–125	800–1000	0.15–0.25	0.5–1	Longit total final 80 mm (o según plano)	
2.2	**Punteado (centrado) cara opuesta**	Broca de centrar (Ø4–5 mm)	15–20	400–600	Manual / 0.05–0.10	3–5 mm total	Avanz con contra asegu perpe refrige	

Nº Op.	Operación	Herramienta	Vc (m/min)	N (rpm)	a(mm/rev)	P (mm)	Notas / Medic clave	Croquis
2.3	Cilindrar Ø28 (desbaste)	Plaquette desbaste	90–120	700–900	0.3–0.5	2–3	Longit aprox. 34 mm + marge	
2.4	Cilindrar Ø28 (acabado)	Plaquette acabado	120–150	1000–1200	0.15–0.20	0.3	Ø28 mm	
2.5	Cilindrar Ø24 (desbaste)	Plaquette desbaste	80–110	800–1000	0.25–0.4	2	Longit aprox. 26 mm	
2.6	Cilindrar Ø24 (acabado)	Plaquette acabado	110–140	1100–1300	0.12–0.20	0.3	Ø24 mm	
2.7	Cilindrar Ø20 (desbaste)	Plaquette desbaste	80–100	900–1100	0.25–0.35	1.5–2	Longit aprox. 16 mm	
2.8	Cilindrar Ø20 (acabado)	Plaquette acabado	100–130	1200–1500	0.10–0.18	0.3	Ø20 mm	
2.9	Chaflanes / redondeos	Herramienta chaflán	80–110	800–1000	manual	—	Todos los escalo	
2.10	Control dimensional final	Calibre, micrómetro, etc.	—	—	—	—	Verific todas medid + conce	